

Group Lab Assignment: **Python Snake Game** in Python

Objective: The objective of this lab is to develop a Python Snake Game using Pygame, focusing on Object-Oriented Programming (OOP), inheritance, and polymorphism. Students will enhance the game by implementing various features and explore how OOP principles can be applied in game development.

Table of Contents

1. Set up.....	1
2. Create, Understand and Run the Python Script	2
3. Add More Fruits	3
4. Add More Sound Effects	4
6. Enhance/Adjust the Score System	4
7. Add Obstacles	4
8. Extra Credit (Each part is 5 points, 15 points in total)	5
Appendix 1: Prompt Engineering and LLM usage	5

1. Set up

- **Create a folder named `python_game`.**
- **open and select it in VSC.**
- **Inside this folder, create two subfolders: `images` and `sounds`.**
- **Download Assets attached to the lab.**

Download the following images and place them in the images folder:

```
python_game /
├── main.py
├── images/
│   ├── apple.jpg
│   ├── head.jpg
│   ├── snake_part.png
│   ├── bomb.jpg
│   ├── orange.jpg
│   └── background. apple.jpg
```

Download the following sound files and place them in the sounds folder:

```
└── sounds/
    ├── eat_sound.wav
    ├── explosion.wav
    └── wall.wav
```

- **In upper menu choose Terminal -> new Terminal**

Task 1. Submit screenshot of your result to prove that Task 1 was done and fully completed.

2. Create, Understand and Run the Python Script

- Create a file named `main.py` in the folder you created.
- Copy the provided code into `main.py`.
Download the following py files and place it in the main folder:
`main.py`
- Open `main.py` and read through the entire script.
- Identify the different classes and functions used in the game.
- Pay attention to how inheritance and polymorphism are implemented with the `Fruit`, `Apple`, `Orange`, and `Bomb` classes.
- Run the Game using `python main.py` command.

High-view Code Explanation

#Imports

```
import pygame
import random
import os
```

Initialize Pygame

```
pygame.init()
```

Screen dimensions

```
SCREEN_WIDTH = 800
SCREEN_HEIGHT = 600
```

Colors

```
WHITE = (255, 255, 255)
BLACK = (0, 0, 0)
```

Get the absolute path of the current directory

```
base_path = os.path.dirname(os.path.abspath(__file__))
```

Game Class

#Initializes the game screen, clock, snake, fruits, and background.

#Loads sound effects and sets the background image's opacity.

__init__ Method:

Spawns three fruits, ensuring at least one bomb is generated.

spawn_fruits Method:

#Main game loop that handles events, moves the snake, checks for collisions, and updates the display.

run() Method:

#Draws the walls around the game screen.

draw_walls Method:

#Displays the current score on the screen.

display_score Method:

Snake Class

#Initializes the snake's size, body, direction, and images for the head and body segments.

__init__ Method:

#Changes the direction of the snake based on user input.

change_direction Method:

#Moves the snake in the current direction and handles the growth of the snake.

move Method:

#Checks if the snake's head has collided with a given position.

check_collision Method:

#Checks if the snake's head has collided with the walls.

check_wall_collision Method:

draw Method:

Draws the snake on the screen.

Fruit Class (Base Class)

#Initializes the fruit's image, position, and point value.

__init__ Method:

#Generates a random position for the fruit.

random_position Method:

reposition Method:

#Draws the fruit on the screen.

draw Method:

#Inherit from the Fruit class and represent specific types of fruits with unique point values and images.

Apple, Orange, Bomb Classes (Derived Classes)

Ask LLMs for more details.

Task 2. Submit screenshots of your result to prove that Task 2 was done and fully completed.

3. Add More Fruits

- Create a new fruit class named Plum that inherits from Fruit.
- Set the image to plum.jpg and the point value to 15.
- Ensure plum.jpg is placed in the images folder.

- Adjust game rules and other parts of the code if appropriate.
- Test game to ensure that the game still works and plums are also generated.

Task 3. Submit screenshots of your result to prove that Task 3 was done and fully completed. Include a screenshot of the code you added / modified.

4. Add More Sound Effects

- Add a new.wav sound effect for when the snake eats a plum.
- Ensure the sound file is placed in the sounds folder.
- Modify the game logic to play this sound when a plum is eaten.

Task 4. Submit screenshots of your result to prove that Task 4 was done and fully completed. Include a screenshot of the code you added / modified.

5. Customize the Background

- Add an option to change the background image dynamically during the game.
- Generate a new Background(s) with the help of LLMs (ChatGPT-4o or DALL-E)
- Make new background file is placed in the right folder.
- Allow the player to press a key (e.g., B) to change the background / cycle through different background images.

Task 5. Submit screenshots of your result to prove that Task 5 was done and fully completed. Include a screenshot of the code you added / modified.

6. Enhance/Adjust the Score System

- Display the highest score achieved during the session.
- Save the highest score to a file and load it when the game starts.

Task 6. Submit screenshots of your result to prove that Task 6 was done and fully completed. Include a screenshot of the code you added / modified.

7. Add Obstacles

- Create an Obstacle class that represents static obstacles on the screen.
- Modify the game logic to place obstacles randomly on the screen.
- Ensure the snake cannot pass through obstacles and loses the game if it collides with one.

Task 7. Submit screenshots of your result to prove that Task 7 was done and fully completed. Include a screenshot of the code you added / modified. Submit all your source code as a .txt file so I can run it if needed.

8. Extra Credit (Each part is 5 points, 15 points in total)

- Add a start screen with game instructions.
- Add a game over screen that displays the final score and a prompt to restart the game.
- Implement a power-up fruit that temporarily increases the snake's speed. Create a PowerUp class that inherits from Fruit. Modify the game logic to handle the effects of the power-up.

Task 8. Submit screenshots of your result to prove that Task 8 - Extra Credit was done and fully completed. Include a screenshot of the code you added / modified. Submit all your source code as a .txt file so I can run it if needed.

Appendix 1: Prompt Engineering and LLM usage

It is allowed to use LLMs during this assignment.

You can refer to the following conversation for the reference, can reproduce it if appropriate (this part is optional) to learn and practice how to code with ChatGPT.

Hi! I want to create a python game in python but at the same time be able to teach my students oop, inheritance and polymorphism. I actually want something like a snake game but with the real image of apple and a real image of python and the python should grow after eating apple

GPT

this is what I have can you adjust so snake can be extendable and can go straight or up or down.
[Provided Snake Images created with DALL-E]

GPT

Uploaded image

here is what I have in my images. adjust accordingly.

GPT

```
\main.py", line 1, in <module>
```

```
    from game import Game
```

```
ModuleNotFoundError: No module named 'game'
```

GPT

I defnetely imtalled it import pygame

```
ModuleNotFoundError: No module named 'pygame'
```

GPT

it works lets add a frame like a wall so if snake hits the wall its game over, more fruits with inheritance and polymorphism, game should have a name python game in python and there

should be score system with various fruits worth various amount of points. also, there should be sounds and some sort of bomb so once snake hits it it explodes and also with sound

GPT

can we add a background image>

GPT

where can I get such sounds/

GPT

make it all one big script.

GPT

good this works for some reason I always had an apple generated and never an orange.

GPT

good so I did not have a single bomb generated. also, when the snake hit a wall there should be wall sound

GPT

I want background with opacity.

GPT

nothing happened after hitting the wall but bomb worked well

GPT